THE LOCALIZATION AT A CLOSED SUBSET OF THE FORM V(F_1, ..., F_R) OF A W-CONTRACTIBLE RING IS W-CONTRACTIBLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the construction of the auxiliary faithfully flat weakly etale map to a product of localizations and B, the retraction obtained from w-contractibility of A, and the localization of its image in the intended component through the specialization property.

Editorial note. The label lemma-localization-w-contractible names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-localization-w-contractible –start 2282 –end 2305 –local-id 28004 returned RESULT=OK (L28004 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The localization construction attached to a closed subset. Same chapter, the lemma producing the ring A_Z^tilde whose spectrum consists of the points specializing to the closed subset Z. Supplies the ring B = A_Z^tilde and the fact that every point of its spectrum specializes into Z.
- Faithful flatness and weak etaleness of products of localizations. Algebra, the basic properties of localizations and products giving that the displayed product map is faithfully flat and weakly etale. Supplies the hypothesis to which w-contractibility of A is applied.

Source reference. Original reference 096V: sources/stacks-proetale/pinned-original-source.tex, source lines 710–710. Prerequisite pointer invoked at proof line(s) 2286; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let A be a ring. Let $Z \subset \mathrm{Spec}(A)$ be a closed subset of the form $Z = V(f_1, \ldots, f_r)$. Set $B = A^\sim_Z$, see Lemma 096V. If A is w-contractible, so is B.*

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Proof. Let $A_Z^\sim \to B$ be a weakly étale faithfully flat ring map. Consider the ring map

$$A \longrightarrow A_{f_1} \times \ldots \times A_{f_r} \times B$$

this is faithful flat and weakly étale. If A is w-contractible, then there is a retraction σ. Consider the morphism

$$\mathrm{Spec}(A_Z^\sim) \to \mathrm{Spec}(A) \xrightarrow{\mathrm{Spec}(\sigma)} \coprod \mathrm{Spec}(A_{f_i}) \amalg \mathrm{Spec}(B)$$

Every point of $Z \subset \mathrm{Spec}(A_Z^\sim)$ maps into the component $\mathrm{Spec}(B)$. Since every point of $\mathrm{Spec}(A_Z^\sim)$ specializes to a point of Z we find a morphism $\mathrm{Spec}(A_Z^\sim) \to \mathrm{Spec}(B)$ as desired. $\square$